

1. Let S denote the memoryless (probabilistic) binary erasure channel with input alphabet $F = \{0,1\}$ and output alphabet $\Phi = F \cup \{?\}$ and erasure probability $p = 0.1$. A binary simple parity check code $\mathcal{C}$ of length 4 is transmitted through S and the following decoder $\mathcal{D} : \Phi^4 \rightarrow \mathcal{C} \cup \{e\}$ is applied to the received vector:

$$\mathcal{D}(\underline{y}) = \begin{cases} \underline{c} & \text{if } \underline{y} \text{ agrees with exactly one codeword } \underline{c} \in \mathcal{C} \text{ on unerased locations} \\ "e" & \text{otherwise} \end{cases}$$

Compute the probability that output of $\mathcal{D}$ is "e". (5 marks)

2. Consider an adversarial binary erasure channel which can erase t_e positions adversarially. Consider an $(n, k, d_{\min})$ code. What is the maximum value of t_e that the code can tolerate? Assume the decoder given in the previous question. Please give justification. (3 marks)
3. Let $\mathcal{C}$ be an (n, M, d) code over $\mathbb{F}_2$ and that τ and σ be non-negative integers such that $2\tau + \sigma \leq d - 1$. Show that there is a decoder $\mathcal{D} : \mathbb{F}_2^n \rightarrow \mathcal{C} \cup \{"e"\}$ with the following properties:
- If number of errors is τ or less, then errors will be recovered correctly.
 - Otherwise if the number of errors is $\tau + \sigma$ or less, then they will be detected.

(5 marks)

4. Consider an $(n, k, d_{\min})$ binary linear block code with generator matrix G . Let s denote the minimum integer such that any s columns of G have rank k . Then, prove that $s = n - d_{\min} + 1$. (6 marks)

5. Let $G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$ be the generator matrix of a code.

- (a) What is the minimum distance of this code? (2 marks)
- (b) What is the maximum number of errors which can be detected by the code t_d and what is the maximum number of errors which can be corrected by this code t_c ? (2 marks)
- (c) Find the parity-check matrix of the code. (2 marks)